Practical 8: Sorting and Searching Algorithms (CO2)

Code:

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr

def binary_search(arr, target):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1

data = [64, 25, 12, 22, 11]
sorted_data = bubble_sort(data.copy())
print("Sorted Array:", sorted_data)
print("Index of 22:", binary_search(sorted_data, 22))
```

Output:

```
Sorted Array: [11, 12, 22, 25, 64]
Index of 22: 2
```